RouteRec: Behavior-Guided Sparse Routing for Sequential Recommendation

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Abstract

In sessionized sequential recommendation, sessions are short but cross-session history is available. This brevity sharpens behavioral differences across sessions: some sessions are repeat-heavy, others fast and exploratory, others narrowly focused. Existing models use this behavioral context to enrich the sequence representation — improving *what the model encodes*. We ask whether behavioral context should instead determine *which computation path* the model applies, and what observable signal should drive that routing decision.

RouteRec pursues this direction with Mixture of Experts (MoE). The central design choice is the routing input. Hidden states summarize the sequence but are optimized for prediction, not for encoding behavioral regime, and they confound representation with routing control in the same latent vector. RouteRec instead computes *behavioral cues* — compact statistics derived directly from interaction logs — and routes on them through a control path kept strictly separate from the sequential backbone. This separation makes the routing input explicit, grounds route choices in observable quantities, and keeps representation and routing control from interfering. RouteRec extracts cues at macro, mid, and micro temporal scopes and applies hierarchical sparse expert allocation to concentrate computation on a small, focused expert subset.

In the main seen-target comparison across six datasets under a shared sessionized temporal protocol, RouteRec achieves the best overall Avg. Rank of 1.67, versus 3.56 for the next best baseline. Gains are clearest on Foursquare, LastFM, and KuaiRec, and diagnostics confirm more behavior-consistent expert selection. Anonymous code: https://anonymous.4open.science/r/RouteRec/.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, sparse routing, conditional computation, behavioral cues, mixture of experts

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1 Introduction

Session-based sequential recommendation models next-item prediction from interactions divided into temporally bounded sessions [16]. Sessions are short, and this brevity makes the structural character of each session particularly pronounced: a session built on rapid revisits to familiar items looks structurally different from one that sweeps across diverse categories in quick succession, even when both users share similar long-run preferences. These differences — in interaction pace, repetition, category focus, and popularity exposure — are directly observable from interaction logs and vary consistently across users and time [42].

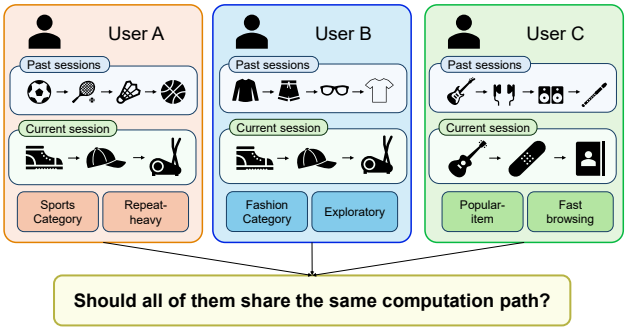

Figure 1: Motivating example: sessions with different structural character can still receive the same shared computation.

Figure 1 illustrates the central gap: sessions with clearly different structural character can still pass through the same shared transformation. Existing work has addressed behavioral heterogeneity primarily by enriching the sequence representation. Time-aware self-attention [25] embeds interval structure into the backbone. Contrastive objectives [32] and frequency-aware inductive biases [7, 38] shape the representation geometry. Feature-aware models [45, 50] incorporate side information into the encoding path, and cross-session context has been brought in through hierarchical architectures [33] and session-aware extensions [1]. These approaches improve *what the model represents*. They still apply the same transformation to all prefixes, regardless of structural differences across sessions. Whether behavioral context should also determine *which transformation the model applies* — rather than only enriching the representation that a single shared block processes — is a question these approaches do not address.

Conditional computation — executing only a data-dependent subset of model parameters — makes this question concrete. We ask: in sessionized sequential recommendation with accessible cross-session history, can explicit behavioral cues from interaction logs serve as the control signal for *which computation path* the model applies, rather

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than only as inputs for representation enrichment? We do not treat a different computation path as a necessary requirement. The question is a design one: *given conditional computation, what observable signal should drive routing?*

Mixture of Experts (MoE) [15, 18] provides the mechanism: a router that selects which subset of expert modules processes each input, rather than passing all inputs through the same shared transformation. Sparse MoE [6, 8, 37, 52] makes that selection efficient by activating only a small subset at a time. Once MoE is introduced, the routing input becomes a substantive design choice. The natural default is to route from hidden states, since they summarize the sequence. But hidden states are optimized for prediction, not for encoding the behavioral regime of a session — its observable structural character, such as repeat-heavy consumption or fast exploratory browsing. Routing from hidden states also confounds two roles in the same vector: the quantity that drives expert selection is identical to the quantity being shaped by that selection, so the behavioral basis of any given route remains implicit. In short sessions, this is further compounded because the hidden state may not yet reflect cross-session behavioral character when routing decisions are made.

A separate routing control path avoids this confound. RouteRec keeps two channels distinct: the sequential backbone encodes the sequence for next-item prediction, while a dedicated routing control path computes compact scalar summaries — *behavioral cues* — directly from interaction logs and uses them to select experts. Behavioral cues are organized into four families that cover the main observable dimensions of session structure: Tempo (interaction pace and interval dynamics), Focus (category concentration and switching), Memory (repetition and recurrence patterns), and Exposure (popularity level and drift). Because cues are observable statistics rather than learned latent values, route changes can be traced back to specific behavioral quantities visible in the log — an advantage of the separation, not an independent design claim. Cues are computed at three temporal scopes because cross-session history, within-session progression, and short-term local transitions each expose different kinds of behavioral evidence.

Instantiating this design raises three concrete tensions. Behavioral cues must remain portable across datasets where only item identities, timestamps, and sometimes coarse category labels are available, yet still be informative enough to differentiate computation paths. Routing decisions must be separated by temporal scope so that slow cross-session tendency and fast local dynamics each get a distinct control view. And expert selection must be sparse enough to produce real routing differences across sessions, rather than collapsing into a soft shared mixture that differs little from a shared transformation.

RouteRec addresses each tension directly. It constructs behavioral cues from the minimal signals present in ordinary interaction logs, organizes routing across macro, mid, and micro temporal scopes, and uses hierarchical sparse expert allocation to enforce *route commitment* — the property that routing concentrates on a small, focused expert subset rather than spreading weight broadly. The sequential backbone remains intact for prediction; the computation path is governed separately by observable behavioral cues.

The main contributions of this work are:

- We identify the routing input as a substantive design decision in sessionized sequential recommendation: once MoE introduces conditional computation, routing from hidden states versus explicit behavioral cues is a choice with direct consequences for routing control, expert specialization, and inspectability.
- We propose RouteRec, which separates a sequential backbone for prediction from a routing control path that reads behavioral cues directly from interaction logs — keeping representation and routing control strictly distinct.
- We instantiate this design with three components — Behavioral Cue Construction, Coarse-to-Fine Multi-Stage Routing, and Hierarchical Sparse Expert Allocation — that together make routing grounded, multi-scope, and committed.
- Empirically, RouteRec achieves the best overall Avg. Rank in the main seen-target comparison across six datasets under a shared sessionized temporal protocol, with the clearest gains on Foursquare, LastFM, and KuaiRec.

2 Related Work

2.1 Session-based and Sequential Recommendation

Sequential recommendation predicts the next item from an ordered interaction history [43]. Session-based recommendation is a closely related setting in which those interactions are partitioned into short temporally bounded sessions and prediction is made from the current session context [16, 42]. Earlier models established the main sequence-modeling backbone families for this problem: FPMC [35] factorized personalized Markov transitions, Caser [41] and NextItNet [48] applied convolutional sequence modeling, and GRU4Rec [13] showed that recurrent models can learn strong session dynamics. NARM [24], STAMP [27], and SR-GNN [44] then emphasized session intent, short-term memory, and transition structure more explicitly.

Transformer-based models such as SASRec [19] and BERT4Rec [39] became strong general backbones for sequential recommendation. More recent work adds stronger inductive biases through contrastive regularization [14, 32], frequency-aware modeling [7, 38], and selective state-space backbones [28]. These models differ substantially in how they encode the sequence. They still apply a shared computation path to all prefixes. For short sessions with visibly different structural character, this leaves open whether the model should adapt only its representation or also its computation path.

2.2 Behavioral Context and Router Inputs

Behavioral heterogeneity across sessions is a recognized issue in the recommendation literature. Hierarchical and session-aware models such as HRNN [33], SHAN [47], and more recent hybrid session architectures [1] separate broader preference context from within-session dynamics. Time-aware self-attention [25] injects interval structure into the backbone. Feature-aware models such as FDSA [50] and DIF-SR [45] incorporate side information to refine the sequence representation. RepeatNet [34] and ICSRec [31] focus on repetition and evolving intent. Multi-interest models such as MIND [23] and ComiRec [5] represent user behavior through multiple latent interests within a single forward pass.

Across these lines of work, behavioral context is used to improve what the model represents — injected into hidden states, attention patterns, or auxiliary branches — while the computation applied to each prefix remains shared. RouteRec asks a different question: whether compact scalar summaries of session structure, derived directly from interaction logs without enriching the representation, should instead control *which* expert subset is activated for each prefix.

2.3 Mixture of Experts in Recommendation

Mixture of Experts (MoE) [15, 18] replaces a shared transformation with a set of expert modules, each activated selectively by a learned router. Sparse MoE [8, 37] makes this conditional computation efficient by activating only a small subset of experts per input. In recommendation, MMoE [29] and PLE [40] use expert decomposition mainly for multi-task learning, where different tasks benefit from different expert subsets. More recent recommendation models such as FAME [26], M3SRec [2], and HM4SR [49] also introduce experts, but routing is typically driven by learned hidden representations or modality-specific states rather than by explicit behavioral cues computed from the interaction log.

Work on sparse MoE mechanics, including expert choice routing [51] and z-loss stabilization [52], improves how sparse activation is executed. It does not determine what the router should read. That choice is especially important in session-based sequential recommendation because routing decisions must often be made from short prefixes whose hidden states need not yet expose cross-session behavioral character clearly. RouteRec targets this gap by defining behavioral cues directly from interaction logs and by keeping the routing control path separate from the representation path.

3 Problem Setting and Design Challenges

3.1 Problem Setup

For each user, let $\mathcal{S}_u = [s_1, \ldots, s_M]$ denote the chronologically ordered session sequence. A session $s_m = [x_{m,1}, \ldots, x_{m,L_m}]$ contains ordered interactions, where each interaction includes at least an item identity and timestamp, and may additionally include a coarse grouping field such as category, genre, or artist. At prediction time, the observed context is the current-session prefix together with earlier sessions. When the session index is clear, we write the current prefix as $x_{1:t}$ and the next target as x_{t+1}.

The task is next-item prediction: given the observed history up to position t in session s_m, the model predicts x_{t+1}. RouteRec uses only signals commonly available after sessionization — item identity, timestamp, optional coarse grouping label, and corpus-level popularity statistics computed from training data. No heavy side-information encoders or auxiliary supervision are required.

3.2 Design Challenges

Using behavioral cues from interaction logs as the explicit routing input — rather than hidden states — raises three design tensions. Each must be resolved for cue-driven routing to produce meaningful computation differences across sessions, rather than simply adding a routing step that leaves behavior largely unchanged.

C1. Portability versus informativeness. Behavioral cues must be computable from whatever signals are reliably present across datasets. In practice, this is narrow: item identities, timestamps, and sometimes a coarse grouping field such as category or genre. A cue derived from this restricted set risks carrying too little behavioral signal to differentiate computation paths; a cue that requires richer side information is unavailable in most benchmark settings. Constructing cues that are both portable and informative enough to support route differences is the first tension.

C2. Scope alignment of behavioral evidence. Behavioral evidence emerges at different temporal ranges. Cross-session repetition and drift are visible only over multiple past sessions; within-session category concentration builds across a few minutes of activity; local transition patterns appear only in the last few interactions before the prediction target. A single routing view that draws on all these ranges simultaneously produces a cue that conflates slow structural tendency with fast local dynamics, and cannot respond distinctly to either. Routing decisions must be separated by the temporal scope at which each kind of evidence becomes a reliable signal.

C3. Route commitment versus coverage. Even with explicit cue-driven routing, behavior-dependent computation collapses if routing spreads weight broadly across experts. The model then reduces to a soft shared mixture regardless of the cue: a repeat-heavy session and a fast-exploratory one are routed almost identically, and the value of cue-driven routing is lost. Expert selection must be sparse enough to concentrate computation on a small, focused subset — producing real routing differences across sessions — while retaining enough coverage to handle the full range of behavioral regimes observed in practice.

The following section addresses each tension: behavioral cue construction (C1), multi-stage routing separated by temporal scope (C2), and hierarchical sparse expert allocation (C3).

4 Method

4.1 RouteRec Overview

Figure 2 summarizes RouteRec at the framework level. RouteRec retains a strong sequential backbone for sequence encoding, but replaces shared feed-forward computation with behavior-guided routing stages. The model follows the two-path design motivated in Section 1: hidden states remain the representation channel for prediction, while a separate routing control path computes behavioral cues from observable history and uses them to select experts.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the routing-stage set. These stages correspond to broad cross-session, within-session, and local evidence scopes. Let $\mathbf{h}_t^{\text{in},(s)}$ denote the hidden state entering stage $s \in \mathcal{T}$, and let $\mathbf{f}_t^{(s)}$ denote the cue vector supplied to that stage. The common routing interface is

$$\boldsymbol{\pi}_t^{(s)} = \text{softmax}\Big(\text{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{1}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \, \text{Expert}_e^{(s)}(\mathbf{h}_t^{\text{in},(s)}), \tag{2}$$

where E is the number of experts at stage s, τ_s is a routing temperature, and $\boldsymbol{\pi}_t^{(s)}$ is the expert-weight vector. This interface is shared across stages. What changes across stages is the behavioral evidence used for routing and the way the routed computation is organized.

The rest of the method turns the design challenges in Section 3 into a concrete model. Section 4.2 constructs behavioral cues from interaction logs. Section 4.3 organizes routing across macro, mid, and micro scopes. Section 4.4 defines hierarchical sparse expert allocation so that routing stays focused on a small expert subset rather than diffusing broadly. In the main configuration, only a limited number of expert groups and experts is active at each routing decision.

We first present RouteRec as a design framework and refer to the main experimental configuration only when implementation details matter. Alternative variants and broader structural sweeps are deferred to the appendix.

4.2 Route-Guiding Behavioral Cues

To make route guidance available under realistic data conditions, RouteRec uses signals commonly present in interaction logs: item identity, timestamp, optional coarse item-group labels, and corpus-level popularity statistics. These cues are not meant to be optimal predictive features on their own. Their role is narrower: they provide compact observable proxies for when behavior warrants a different computation path. We therefore avoid heavy side-information encoders in the routing branch and instead use compact scalar summaries that remain easy to compute and inspect.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{3}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{5}$$

where $H = 5$ and $w = 5$ in our main setting. These are the default paper windows used throughout the main experiments, and their role is revisited in the structural ablations in Appendix D. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures coverage, pace, and interval dynamics. Focus captures concentration and switching behavior over the coarsest item-group proxy available in the dataset, such as product category, venue category, artist grouping, or genre. Memory captures repetition, overlap, and recurrence structure. Exposure captures the level, spread, and drift of interacted popularity. In the main configuration, this family template yields a compact 16-dimensional cue bank per stage. When such grouping fields are unavailable, the corresponding focus cues are handled by the fixed missing-signal rule documented in Appendix A. Representative examples are listed in Table 4.

The template is shared across stages, but each stage reads it through a different temporal lens: macro reflects persistent tendency, mid tracks evolving within-session intent, and micro emphasizes short-horizon cues near the prediction target. The goal is not exhaustive feature engineering, but a small semantically legible cue bank that serves as a stable routing interface across datasets. Appendix A records the representative cue definitions and main paper setting.

4.3 Coarse-to-Fine Multi-Stage Routing

To keep routing aligned with temporal scope, the stages are not redundant even though the router formula is shared. They differ in temporal evidence, routing granularity, and placement inside the backbone. In the main experimental configuration, the serial layout is

$$\underbrace{\text{self-attn}}_{\text{context}} \rightarrow \text{macro} \rightarrow \text{mid} \rightarrow \underbrace{\text{self-attn}}_{\text{refresh}} \rightarrow \text{micro}.$$

Let $\mathbf{h}_t^{(0)}$ denote the hidden state after the first attention block, $\mathbf{h}_t^{(1)}$ after macro routing, $\mathbf{h}_t^{(2)}$ after mid routing, and $\mathbf{h}_t^{(3)}$ after the second attention block. Macro, mid, and micro therefore consume $\mathbf{h}_t^{(0)}$, $\mathbf{h}_t^{(1)}$, and $\mathbf{h}_t^{(3)}$, respectively. The order is deliberate: the first attention block provides a contextualized sequence state for the slower macro and mid routing decisions, whereas the second refreshes token interactions immediately before the local micro router. Appendix D reports the broader layout ablations.

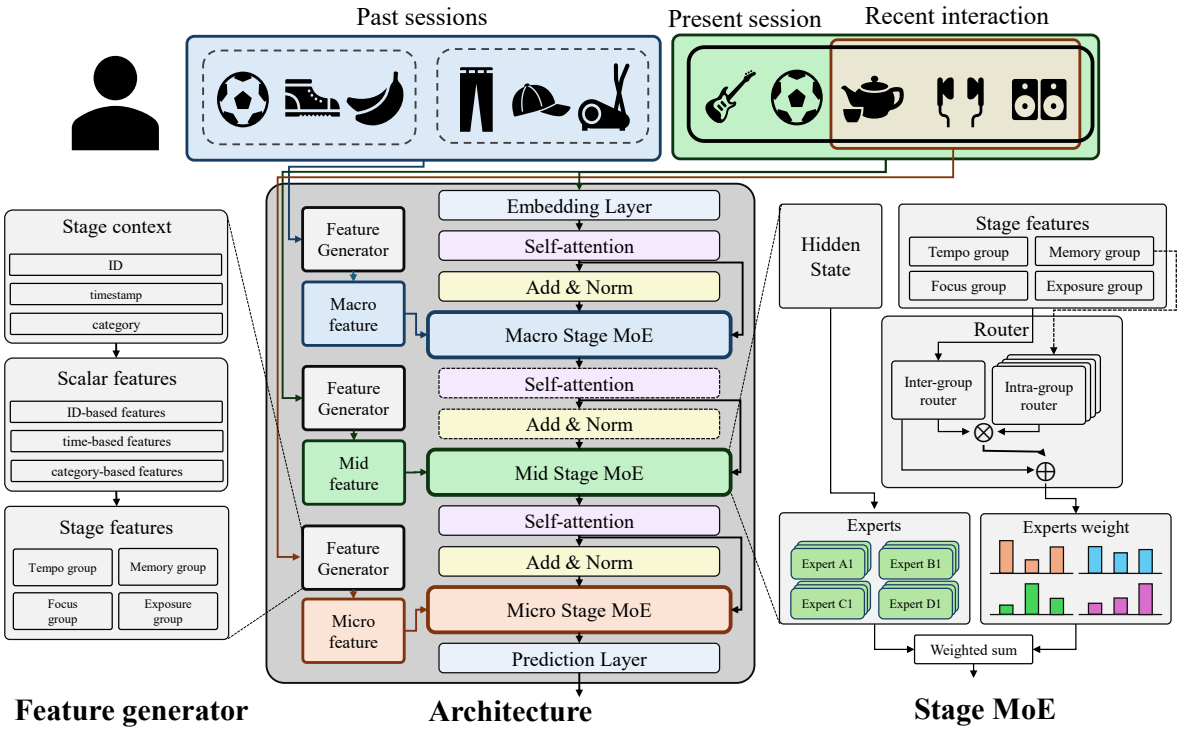


Figure 2: Overall RouteRec framework. Read left to right: panel (a) shows stage-specific cue extraction, panel (b) shows the serial backbone, and panel (c) shows hierarchical sparse expert selection. RouteRec keeps a strong sequential backbone and replaces shared feed-forward blocks with behavior-guided sparse routing stages. All stages use the same hierarchical sparse routing principle: group-level scores select a subset of semantic groups, then conditional intra-group scores select a subset of experts inside each chosen group before forming the active expert mixture. Macro, mid, and micro differ by temporal evidence scope and routing granularity, not by the basic sparse routing logic. This is a framework-level view; the exact serial configuration used in the main experiments is specified in Section 4.3 and Appendix A.

Macro and mid (sequence-wise routing). Macro and mid compute one routing distribution per sequence from their respective windows, then broadcast it over valid positions inside the sequence. Macro uses broader cross-session history; mid uses within-session prefix evidence. This setting emphasizes stable allocation for slower-changing context, where token-wise routing tended to add variance without introducing an equally meaningful control resolution in our ablations.

Micro (position-wise routing). Micro computes routing per token near the prediction target using short-horizon context. This enables fast adaptation to immediate transitions without changing the router form itself.

Formally, let $\bar{\mathbf{f}}_i^{(s)}$ denote the sequence-level cue summary used for sequence-wise routing at stage s. Then macro and mid use

$$\bar{\boldsymbol{\pi}}_i^{(s)} = \mathrm{softmax}\Big(\mathrm{Router}_s(\bar{\mathbf{f}}_i^{(s)})/\tau_s\Big), \qquad s \in \{\mathrm{macro}, \mathrm{mid}\},$$

$$\boldsymbol{\pi}_{i,t}^{(s)} = \bar{\boldsymbol{\pi}}_i^{(s)}.$$

Micro instead applies the same router form directly to position-specific cues,

$$\boldsymbol{\pi}_{i,t}^{(\mathrm{micro})} = \mathrm{softmax}\Big(\mathrm{Router}_{\mathrm{micro}}(\mathbf{f}_{i,t}^{(\mathrm{micro})})/\tau_{\mathrm{micro}}\Big).$$

Hence, RouteRec's coarse-to-fine behavior comes from stage evidence scope and update frequency, while the underlying router composition remains constant. The stages are therefore not repeated copies of the same router; they correspond to different temporal horizons at which behavioral evidence becomes informative for computation allocation. Because each stage captures a distinct control view, the final model uses sparse route commitment so that only a small subset of stage-relevant groups and experts is activated at each decision.

4.4 Hierarchical Sparse Expert Allocation

To structure expert selection rather than merely enlarge capacity, the final RouteRec architecture uses one shared hierarchical sparse composition at every stage. Experts are partitioned into G groups with C experts per group, so

$E = GC$. For stage s and position t, the router produces

$$p_{t,g}^{(s)} = \mathrm{softmax}_g\left(z_{t,g}^{(s)}\right), \tag{6}$$

$$r_{t,g,c}^{(s)} = \mathrm{softmax}_c\left(u_{t,g,c}^{(s)}\right), \tag{7}$$

where $p_{t,g}^{(s)}$ is a group-level routing distribution and $r_{t,g,c}^{(s)}$ is a conditional within-group routing distribution. The final expert probability is their product:

$$\pi_{t,g,c}^{(s)} = p_{t,g}^{(s)}\, r_{t,g,c}^{(s)}, \tag{8}$$

followed by flattening (g, c) to expert index e. Eq. (8) is the same for macro, mid, and micro at the score level.

In the main configuration, groups are aligned with the four cue families, while multiple experts expand capacity within each group. This is a router-level semantic prior rather than a hard hand assignment of individual cues to experts. All three stages use the same feature-driven group-conditional router: a group-level router selects the relevant semantic control view, and a within-group router allocates capacity inside that view.

The final paper configuration adopts sparse activation at both routing levels. For compactness, let $\mathrm{TopK}(\mathbf{v}, k)$ denote the index set of the k largest entries of a vector $\mathbf{v}$. We first keep only the top-k_g groups,

$$\bar{p}_{t,g}^{(s)} = \frac{p_{t,g}^{(s)}\, \mathbf{1}[g \in \mathrm{TopK}(\mathbf{p}_t^{(s)}, k_g)]}{\sum_{g'} p_{t,g'}^{(s)}\, \mathbf{1}[g' \in \mathrm{TopK}(\mathbf{p}_t^{(s)}, k_g)]}, \tag{9}$$

and then, inside each retained group, only the top-k_e experts,

$$\bar{r}_{t,g,c}^{(s)} = \frac{r_{t,g,c}^{(s)}\, \mathbf{1}[c \in \mathrm{TopK}(\mathbf{r}_{t,g,:}^{(s)}, k_e)]}{\sum_{c'} r_{t,g,c'}^{(s)}\, \mathbf{1}[c' \in \mathrm{TopK}(\mathbf{r}_{t,g,:}^{(s)}, k_e)]}. \tag{10}$$

The executed expert route is then

$$\hat{\pi}_{t,g,c}^{(s)} = \bar{p}_{t,g}^{(s)}\, \bar{r}_{t,g,c}^{(s)}, \tag{11}$$

and the stage output uses only the active experts:

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{g=1}^{G} \sum_{c=1}^{C} \hat{\pi}_{t,g,c}^{(s)}\, \mathrm{Expert}_{g,c}^{(s)}(\mathbf{h}_t^{\mathrm{in},(s)}). \tag{12}$$

In the main experimental configuration, $G = 4$ cue-aligned groups and $C = 3$ experts per group. We choose $k_g = 3$ and $k_e = 2$, so each routing decision activates $3 \times 2 = 6$ experts out of 12 total experts. The semantic group prior therefore remains soft at the score level, but the executed computation path is explicitly sparse.

Sparse routing is introduced not only to reduce computation but also to enforce route commitment and sharpen specialization around a small active set. The hierarchical factorization keeps that selection interpretable by organizing it around semantic control views, while the experts themselves remain fully learned. Dense soft mixing is treated as an ablation variant rather than the final choice.

4.5 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers in the main paper configuration: route consistency and z-loss.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\mathrm{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. We also form a stage-level cue summary

$$\bar{\mathbf{f}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \mathbf{f}_{i,t}^{\,s}.$$

For each stage s, we construct a pair set $\mathcal{P}_s$ from the k_{nn} nearest non-self samples in the mini-batch according to cosine similarity between these stage-level cue summaries, and minimize

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \mathrm{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big),$$

which encourages neighboring behavioral states to produce compatible routing distributions. Appendix A reports the neighbor setting used in the main configuration. To stabilize gating numerics, we adopt z-loss on the pre-softmax expert logits $a_{i,t,e}^{s}$ [52]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp a_{i,t,e}^{s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}}\, \mathcal{L}_{\mathrm{cons}} + \lambda_z\, \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. Unlike classical MoE setups that explicitly enforce uniform expert traffic, the main configuration does not include load balancing. The aim is stable, behavior-aligned specialization rather than equal usage for its own sake. Appendix F reports the objective variants.

5 Experiments

5.1 Experimental Setup

We evaluate RouteRec on six public datasets spanning e-commerce, POI recommendation, short-video consumption, music listening, movie preference, and anonymous clickstream behavior: Beauty, Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket. Appendix B summarizes their provenance and processed statistics.

Across datasets, we use a sessionized temporal protocol assembled from established session-based and session-aware practice rather than a canonical raw-sequence leaderboard setting [16, 22, 33]. Interactions are ordered chronologically, sessionized with inactivity heuristics, and iteratively filtered to remove sessions shorter than 5 and items with fewer than 3 interactions [24, 44]. We use a 30-minute threshold for

Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket, and a 14-day threshold for the much sparser Beauty review stream [3, 20, 33]. KuaiRec and LastFM are reported on processed sampled subsets (20% and 3%, respectively). After preprocessing, sessions are sorted by start time and split chronologically into 70%/15%/15% train/validation/test partitions. Appendix B gives the full preprocessing and held-out cleanup details [11, 17].

Our main comparison uses the seen-target view, since ID-based recommendation models cannot rank items absent from the training vocabulary and session-aware benchmarks typically treat unseen targets as supplementary accounting rather than the primary comparison [22, 24, 44]. We compare against GRU4Rec, SASRec, TiSASRec, DuoRec, FEARec, BSARec, FAME, DIF-SR, and FDSA [7, 13, 19, 25, 26, 32, 38, 45, 50]. All methods use the same processed files, the same optimization family, matched early stopping, and the same model-selection rule.

We treat tuning as part of the protocol rather than as a secondary implementation detail. Each model is searched within a fixed bounded grid of shared optimization hyperparameters plus a small number of model-specific controls, and any adaptive search remains restricted to those predeclared candidate sets and the same validation files and stopping rule for all methods [9]. RouteRec uses no extra held-out sessions or external metadata at evaluation time; its routing cues are deterministic summaries of the same observed histories used by every method. Appendix B reports the exact grids, budgets, processed statistics, and full $k \in \{5, 10, 20\}$ tables.

The main experimental section is organized around five questions: **Q1)** ranking quality, **Q2)** routing control, **Q3)** design justification, **Q4)** practicality, and **Q5)** routing semantics. Reduced-cue portability studies and broader structural sweeps are deferred to the appendix.

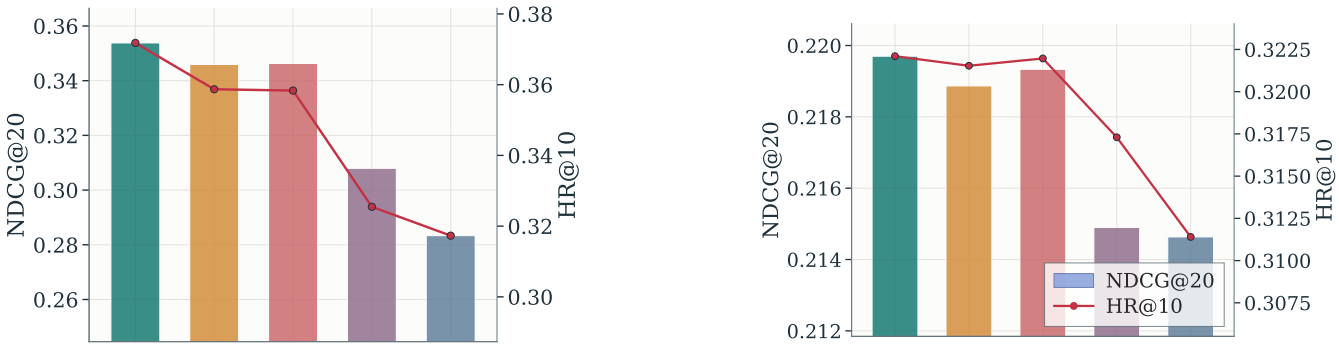
This organization is deliberate for the paper narrative. Q1 establishes whether behavior-guided routing is worth considering at all under a shared sessionized protocol, Q2 and Q3 test whether the gain is actually tied to routing control and staged structure, Q4 checks whether that design remains practical once sparsity is enforced, and Q5 returns to the original semantic claim by asking whether the learned routes vary in the way the method intends.

5.2 Q1. Does RouteRec Improve Ranking Quality?

Table 1 presents the main seen-target comparison in the text using HR@10, NDCG@10, and MRR@20, together with an Avg. Rank summary across all reported rows. This presentation keeps the main paper focused on a compact set of representative cutoffs, while Appendix B reports the full $k \in \{5, 10, 20\}$ grid under the same selection rule. Under this view, RouteRec delivers the strongest overall rank, leads on Foursquare, LastFM, and the sampled KuaiRec subset, remains close to the top on Beauty and Retail Rocket, and

Table 1: Seen-target test results under the sessionized temporal protocol. Datasets marked with † are sampled processed subsets (KuaiRec 20%, LastFM 3%). For each method, we report the seen-target test metrics of the run achieving the highest mean seen-target test score over the nine reported metrics. Avg. Rank is computed over all rows in this table, where lower is better. The RouteRec column is shaded for readability.

Dataset	Metric	SASRec	GRU4Rec	TiSASRec	FEARec	DuoRec	BSARec	FAME	DIF-SR	FDSA	RouteRec
Beauty	HR@10	0.1324	0.0840	0.1235	0.1404	0.1347	0.0726	0.0497	0.1165	0.1318	**0.1561**
	NDCG@10	0.0800	0.0457	0.0831	0.0797	0.0793	0.0504	0.0303	0.0811	0.0918	**0.0948**
	MRR@20	0.0659	0.0361	0.0735	0.0634	0.0662	0.0450	0.0250	0.0716	**0.0822**	0.0794
Retail Rocket	HR@10	0.5321	0.5048	0.5131	0.5487	0.5478	0.4655	0.4612	0.4322	0.5343	**0.5608**
	NDCG@10	0.3907	0.3674	0.3755	0.3998	0.3994	0.3957	0.3943	0.3863	0.3848	**0.4148**
	MRR@20	0.3507	0.3290	0.3368	0.3581	0.3576	**0.3745**	0.3738	0.3709	0.3427	0.3737
Foursquare	HR@10	0.3049	0.2180	0.2976	0.3054	0.3195	0.2383	0.2417	0.3032	0.3042	**0.3215**
	NDCG@10	0.1929	0.1476	0.1926	0.1925	0.1970	0.1609	0.1604	0.2008	0.2014	**0.2042**
	MRR@20	0.1624	0.1286	0.1635	0.1613	0.1631	0.1393	0.1378	0.1723	**0.1726**	0.1715
ML-1M	HR@10	0.1423	0.1456	0.1448	0.1490	0.1555	0.1374	0.1533	0.0669	**0.1606**	0.1568
	NDCG@10	0.0717	0.0778	0.0756	0.0736	0.0810	0.0723	0.0818	0.0331	**0.0833**	0.0807
	MRR@20	0.0555	0.0621	0.0600	0.0559	0.0641	0.0570	0.0651	0.0256	**0.0655**	0.0631
LastFM†	HR@10	0.3775	0.3109	0.3715	0.3876	0.3850	0.3475	0.3359	0.3593	0.3762	**0.3965**
	NDCG@10	0.3097	0.2676	0.3029	0.3187	0.3066	0.3044	0.2972	0.3169	0.3247	**0.3307**
	MRR@20	0.2897	0.2558	0.2831	0.2981	0.2833	0.2915	0.2861	0.3051	0.3103	**0.3113**
KuaiRec†	HR@10	0.3284	0.3061	0.3216	0.3513	0.3468	0.3359	0.3307	0.2333	0.3453	**0.3669**
	NDCG@10	0.3177	0.2589	0.2889	0.3286	0.3215	0.3209	0.3188	0.2229	0.3197	**0.3467**
	MRR@20	0.3150	0.2462	0.2802	0.3227	0.3153	0.3170	0.3160	0.2201	0.3133	**0.3425**
Avg. Rank↓		**6.06**	**8.67**	**6.61**	**4.22**	**4.17**	**6.61**	**6.72**	**6.72**	**3.56**	**1.67**



(a) Dynamic-dataset group: same five controls

(b) Stable/mixed-dataset group: same five controls

Figure 3: Routing-control comparison across two dataset groups. Both panels use the same five variants and differ only in dataset grouping, isolating how the routing-control choice behaves in dynamic versus more stable settings.

stays competitive on ML-1M even when the strongest baseline there remains slightly ahead.

The baseline pattern is also informative. SASRec and Ti-SASRec remain consistently strong under this protocol, which is plausible because the sessionized split produces relatively short, recency-sensitive prefixes where strong autoregressive self-attention and explicit interval modeling already capture much of the available signal [19, 25]. By contrast, BSARec is less competitive here even though it is strong in standard longer-sequence settings: its attentive frequency bias is designed to exploit richer global spectral structure [38], and that advantage is naturally attenuated when sessionization shortens contexts and reduces long-range periodic evidence. DIF-SR also improves less in this setup, which is consistent with its side-information-fusion focus [45]: under our shared protocol, auxiliary features are intentionally lightweight and heterogeneous across datasets, so the side branch contributes less additional structure than in richer feature regimes. These gaps should therefore be read as protocol- and inductive-bias-dependent behavior rather than as evidence of under-tuning, since all methods follow the same bounded search, stopping rule, and processed files.

Overall, Q1 shows that RouteRec is competitive under the shared sessionized protocol, with the clearest gains in the more behaviorally dynamic settings.

5.3 Q2. What Should Control Routing?

Figure 3 compares five variants that separate where behavior information is used: a shared FFN, hidden-only routing, feature fusion by additive bias, mixed hidden+behavior routing, and the final behavior-guided sparse router. Hidden-only routing can improve over a shared FFN, and simple feature reinjection is not entirely ineffective, but neither yields the same consistency as using behavioral cues directly as routing controls. The final behavior-guided design therefore outperforms representation-only feature injection and avoids collapsing the routing input back into hidden-state geometry.

Both panels use the same five variants and differ only in dataset grouping. Panel (a) emphasizes more behaviorally dynamic datasets, whereas Panel (b) reports the more stable or mixed group. Appendix F reports the corresponding objective variants separately so that the routing-signal effect is not conflated with the loss choice.

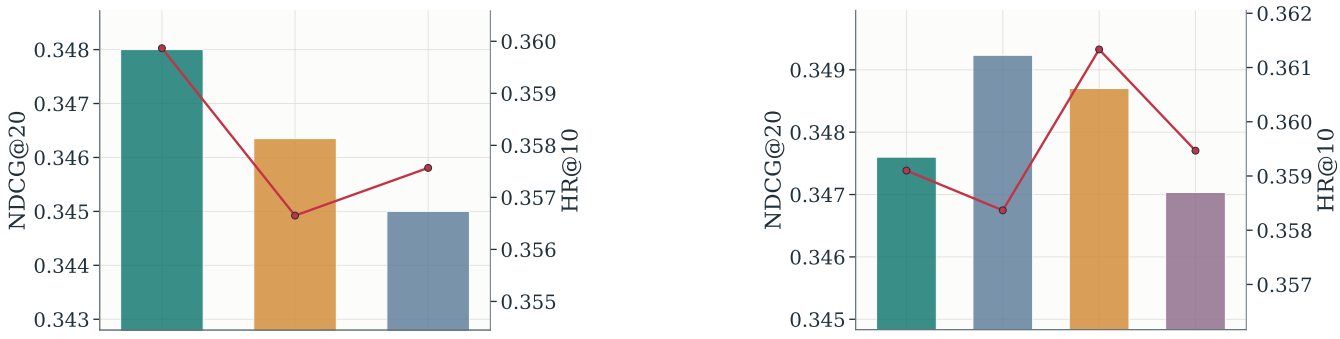
The practical takeaway is that the best routing input is not the one that carries the most raw information, but the one that exposes the cleanest control boundary. Once behavioral cues are used directly at the router, RouteRec can let the representation path focus on sequence modeling while the router absorbs cross-dataset differences in pace, focus, and recurrence more explicitly.

5.4 Q3. Why Is RouteRec's Three-Part Design Effective?

Figure 4 tests whether RouteRec's gains come from the three-part design itself rather than from expert insertion alone. Panel (a) asks whether temporal decomposition is necessary by comparing the final three-stage design against the best reduced single-view and two-view alternatives. Panel (b) asks whether the selected routing organization is meaningful by comparing flat versus hierarchical routing and sparse versus dense allocation under comparable expert pools.

Taken together, these ablations test whether RouteRec's gain comes from temporal decomposition and hierarchical sparse commitment rather than indiscriminate expert insertion. Broader scope-order and portability controls are deferred to the appendix.

The pattern is consistent with the method hypothesis. A single behavior view is too coarse to separate slow tendency from immediate transition pressure, while a dense flat expert pool weakens route commitment even when the total expert budget is comparable. The selected three-stage sparse hierarchy is therefore useful because it assigns different temporal questions to different routing decisions instead of asking one router to absorb all scopes at once.



(a) Temporal decomposition matters (b) Hierarchical sparse routing matters

Figure 4: Design-justification ablations. Panel (a) compares the final three-stage design against reduced temporal decompositions, and Panel (b) compares flat versus hierarchical and dense versus sparse routing organizations.

5.5 Q4. Is RouteRec Practical as a Sparse Design?

Q4 asks whether the final sparse RouteRec design remains practical in cost. On two representative datasets, sparse RouteRec uses 3.2M total parameters but only 1.6M active parameters per routed decision, compared with 3.2M active parameters for the dense variant and 1.8M for SASRec. Its train and inference ratios are 1.42× and 1.18× relative to SASRec, versus 1.85× and 1.38× for dense RouteRec. The key distinction is therefore executed capacity rather than total capacity alone. Appendix G reports the full efficiency summary.

This gap matters for the paper claim because it shows that RouteRec is not simply buying accuracy through unrestricted dense over-parameterization. The sparse design still pays a modest routing overhead, but it keeps the active compute budget much closer to strong shared-path baselines than the dense expertized alternative does.

5.6 Q5. What Does the Learned Router Actually Capture?

Figure 5 addresses the most direct semantic question raised by the method: if RouteRec is designed to react to behavioral regimes, then representative repeat-heavy, fast exploratory, and narrow-focus cases should exhibit different stage-wise routing maps. We therefore present three representative routing cases in the main text and retain intervention-based perturbation results as supporting evidence for the appendix and discussion.

This analysis connects the empirical evidence back to the motivation in Figure 1. To avoid circularity, the cases are selected from simple post hoc statistics derived from raw interaction histories rather than optimized labels or supervision targets. Appendix-level cue interventions support the same reading.

The three cases also clarify why the method is most compelling on dynamic datasets. When repeat structure, short-horizon tempo, or concentration shifts become more pronounced, the stage-local router has a clearer reason to deviate from a shared computation path. In more preference-stable settings, the same mechanism remains competitive but is asked to differentiate less sharply, which is consistent with the smaller margin observed on ML-1M.

Detailed full-cutoff results and extended analyses, including exact sparse-routing specification, reduced-cue portability studies, broader structural ablations, sparse-design controls, objective variants, extended efficiency summaries, routing diagnostics, intervention studies, and transfer experiments, are reported in Appendices A–K.

6 Conclusion

RouteRec starts from a design question: in sessionized sequential recommendation with accessible cross-session history, can behavioral cues from ordinary interaction logs serve as an explicit control signal for conditional computation, rather than only as inputs for representation enrichment? The experiments support an affirmative reading from three angles. First, RouteRec improves next-item ranking most clearly on behaviorally dynamic datasets and remains competitive elsewhere, although the advantage narrows in more preference-stable settings such as ML-1M. Second, the gain is tied to behavioral cues, staged temporal decomposition, and sparse route commitment rather than indiscriminate expert insertion. Third, compact cues derived from ordinary interaction logs preserve a meaningful share of the benefit, keeping the method practical when side information is limited.

Taken together, the results suggest that behavior-guided routing can be both semantically meaningful and computationally selective. In the main configuration, route-guiding cues, coarse-to-fine routing, and hierarchical sparse expert allocation form a compact interface for conditional computation. Future work can extend that interface through adaptive sparsity budgets, transfer of routing priors across domains,

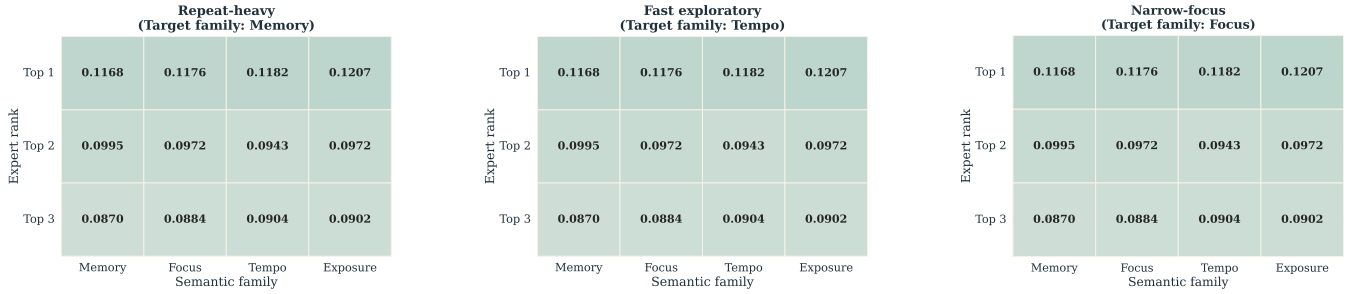


Figure 5: Representative routing semantics. Cases are defined by simple post hoc statistics: repeat-heavy, fast exploratory, and narrow-focus. Each panel shows how stage-wise semantic-group emphasis shifts across regimes.

more robust cue construction, and tighter deployment-aware constraints.

More broadly, the paper suggests a useful modeling stance for sessionized recommendation: not every gain from extra capacity should be interpreted as better sequence representation. Some of it can come from exposing a better control interface over when different computations should be used at all. RouteRec's contribution is to make that interface explicit, lightweight, and interpretable.

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A Exact RouteRec Instantiation and Sparse Routing Formulation

This appendix records the fixed paper instantiation and the representative cue bank used by RouteRec. Whereas the main text emphasizes the conceptual design axes, this section documents the implementation choices that are relevant for reproducibility and for interpreting the later ablations, with the sparse routing formulation made explicit.

A.1 Main Paper Configuration

Table 2: Fixed main configuration used in the paper.

Aspect	Main paper configuration
Backbone layout	self-attn → macro block → mid block → self-attn → micro block
Stage execution	Serial
Routing granularity	Macro and mid: sequence-wise; micro: position-wise
Router composition	Shared hierarchical sparse router at all three stages
Expert activation	Top-3 groups out of 4; top-2 experts per selected group
Active experts	6 out of 12 total per routed decision
History windows	$H = 5$ sessions (macro); $w = 5$ positions (micro)
Core auxiliary terms	Route consistency + z-loss; no load balancing

The placement of the two attention layers reflects the intended temporal roles of the stages. The first attention layer contextualizes the raw prefix before any routing is applied. Macro and mid then operate consecutively because both provide slower sequence-level control over the same contextualized sequence state: macro captures broader recent-history tendency, while mid refines within-session evolution. A second attention layer is inserted only before micro so that the final local router receives a refreshed token interaction state immediately before prediction. This layout was the most stable high-performing architecture in our architecture sweep, and Appendix D compares it against denser and differently ordered alternatives.

This table serves as a reproducibility summary rather than as a search-space description. Architectural choices are fixed across the paper track, whereas capacity and optimization values are selected per dataset from a narrow candidate set and are therefore reported as compact ranges only when no single global value is shared across all reported runs.

A.2 Implementation Specification and Minimal Execution Notes

Table 3: Implementation-level settings used in the selected paper track.

Item	Reported setting
Semantic groups	$G = 4$ groups: Tempo, Focus, Memory, Exposure
Experts per group	$C \in \{2, 3, 4\}$; total experts $E = 4C$
Main sparse setting	$C = 3$, top-3 groups, top-2 experts per selected group
Cue dimensionality	16 scalars per stage
Router hidden width	Representative range 48–128
Routing temperature	Macro 1.0; mid and micro 1.2
Consistency neighbors	$k_{\mathrm{nn}} = 1$ nearest non-self sample per stage
Missing-signal handling	Broadcast when available; zero-fill focus cues if grouping field absent

Minimal execution notes. The implementation uses three simple rules that are easy to miss if one only reads the high-level method description. First, macro and mid cues are computed once per example and then broadcast over the routed positions, whereas micro cues are evaluated at the position level. Second, top-k masking is followed by renormalization inside the active support rather than by leaving the logits in an unnormalized sparse state. Third, if a dataset does not expose the grouping field needed for focus-family cues, the implementation falls back to the fixed missing-signal policy in Table 3 rather than silently redefining the cue family per dataset.

A.3 Representative Behavioral Cue Families

This cue bank is intentionally compact. The feature families are hand-designed in the sense that they encode behavioral hypotheses, but the design goal is to keep them as a small set of summaries that can serve as a stable routing interface across datasets. In practice, the macro bank is instantiated from the recent-session window used by the selected paper instantiation, while mid and micro are computed from the current session prefix at different temporal scopes. The main text therefore reports the family structure and representative examples, while this appendix provides the implementation-oriented view.

B Dataset, Evaluation, and Full Ranking Results

All experiments were conducted on machines equipped with NVIDIA RTX 3090 24 GB GPUs. Each training or evaluation process used a single GPU; multiple GPUs were used only to run independent experiments in parallel, not for multi-GPU training within a single run. This appendix summarizes the six datasets and makes the evaluation protocol explicit, since recent work has shown that session construction and split strategy can materially affect offline recommender results [11, 17]. The original releases are cited here for completeness [4, 10, 12, 30, 36, 46]. The dataset descriptions emphasize domain differences and signal availability because the routing branch depends on which behavioral evidence is exposed in each corpus.

Table 4: Representative cue summaries used by RouteRec in the main configuration. Each stage uses four semantic families with four scalar summaries per family; the table lists representative examples rather than every implementation detail.

Family	Macro (recent-session tendency)	Mid (within-session evolution)	Micro (short-horizon pressure)
Tempo	Context-valid ratio, last inter-session gap, mean pace, pace trend	Valid-prefix ratio, mean/std interaction interval, session age	Last gap, short-window mean gap, short-window std, deviation from mid-scale pace
Focus	Theme entropy, dominant-theme mass, theme repeat rate, theme shift rate	Category entropy, category top-1 mass, category switch rate, category uniqueness	Immediate category switch, mismatch to recent category history, suffix entropy, suffix uniqueness
Memory	Repeat intensity, adjacent category overlap, adjacent item overlap, repeat trend	Item uniqueness, repeat rate, novelty rate, longest item run	Reconsumption flag, suffix reconsumption rate, suffix item uniqueness, suffix longest run
Exposure	Mean popularity, popularity spread, popularity entropy, popularity trend	Mean popularity, popularity std, popularity entropy, popularity drift	Last-item popularity, suffix popularity spread, suffix popularity entropy, deviation from mid-scale popularity

B.1 Datasets and Protocol

Table 5: Datasets used in the experiments. Datasets marked with † are sampled processed subsets.

Dataset	Domain	Interactions	Sessions	Items	Avg. len.
Beauty	E-com.	33,488	4,243	3,625	7.9
Foursquare	POI	145,238	25,369	30,588	5.7
KuaiRec†	Video	287,411	24,458	6,477	11.8
LastFM†	Music	470,408	25,089	52,510	18.8
ML-1M	Movie	575,281	14,539	3,533	39.6
Retail Rocket	Browse	821,243	153,092	90,211	5.4

Table 5 reports the statistics of the final processed datasets used in the experiments rather than the raw corpus sizes. In particular, KuaiRec corresponds to the 20% processed subset used in our runs, and LastFM corresponds to the 3% processed subset.

For reproducibility, the protocol is summarized as follows.

(1) Sort all interactions chronologically within each dataset.
(2) Reconstruct sessions with a dataset-specific inactivity threshold.
(3) Iteratively prune sessions with length below 5 and items with support below 3 until the constraints are simultaneously satisfied.
(4) Sort the resulting sessions by session start time.
(5) Assign the earliest 70% of sessions to train and treat the latest 30% as a held-out tail.
(6) Build validation and test from the held-out tail under chronology-preserving cleanup.
(7) Report seen-target metrics as the primary view and keep unseen-target behavior as supplementary accounting.

Beauty and Retail Rocket stress sparse e-commerce revisit behavior, Foursquare and KuaiRec emphasize stronger short-horizon intent changes, and LastFM and ML-1M provide longer preference-driven histories. This spread makes it possible to examine whether behavior-guided routing is primarily beneficial in highly dynamic settings or remains competitive when preference structure is more stable.

Signal availability also differs in a way that matters for routing. Beauty exposes item, time, and product-category cues after low-rating removal. Foursquare includes item, timestamp, and venue-category structure with bursty check-in dynamics. KuaiRec offers item, time, and coarse content-side metadata in short-video sessions with strong short-term intent shifts. LastFM provides repeated listening histories with artist-side signals, while ML-1M contributes longer preference-driven sequences with genre information. Retail Rocket is the sparsest side-information setting, relying mainly on anonymous event streams with item and time signals.

We position our setup as a sessionized sequential evaluation protocol assembled from established components, not as the canonical raw-sequence protocol for sequential recommendation [16, 22, 33]. Sessions are reconstructed post hoc with inactivity heuristics, which is standard practice when logs do not provide explicit session boundaries [16]. The thresholds are adapted to the logging regime of each corpus: Foursquare, KuaiRec, LastFM, ML-1M, and Retail Rocket use a 30-minute inactivity threshold, consistent with dense-log session-aware practice [20, 33], whereas Beauty uses a 14-day threshold because review timestamps are substantially more diffuse. More generally, this threshold choice reflects the observation that sparse and item-poor domains often require longer or more domain-specific notions of a session than dense clickstream data [3]. After sessionization, we repeatedly remove sessions shorter than 5 and items with fewer than 3 interactions until no further pruning is required. This iterative filtering follows the same general rationale as common session-based benchmark preprocessing, where short sessions and very low-support items are filtered to stabilize next-item supervision and evaluation [24, 44].

Session splits are constructed at the session level after sorting by session start time. The earliest 70% of sessions are assigned to training, and the remaining 30% are treated as a held-out tail. This tail is first divided into validation and test using stratification over three factors: whether the target item already appears in the prefix, the target item's popularity bucket with respect to the training split, and the session-length bucket. For the final evaluation files, we then merge the held-out sessions, restore chronological order, and re-split them contiguously into equal validation and test halves. This chronology-preserving design is meant to reduce

leakage and keep the offline protocol aligned with temporal deployment order [11, 17].

For the primary evaluation view, we report seen-target metrics only. Concretely, in the held-out files, non-target rows whose items are unseen in training are removed, and unseen targets are retained only for supplementary accounting rather than the main comparison. This matches long-standing session-based and session-aware evaluation practice for ID-based recommendation models, which cannot rank targets outside the training vocabulary and therefore typically separate unseen-target behavior from the primary benchmark table [22, 24, 44]. We report HR, NDCG, and MRR at $k \in \{5, 10, 20\}$ for all methods.

For the result tables, each method is represented by the run with the highest mean seen-target test score over the nine reported metrics under the same selection rule used throughout the paper.

All methods use the same processed train/validation/test partitions. RouteRec's additional macro, mid, and micro cues are deterministic summaries derived from those same observed histories, so the comparison should be read as evaluating different ways of exploiting the shared sessionized interaction context rather than different access to held-out information.

B.2 Model Selection and Tuning Protocol

Because this benchmark is built from a sessionized temporal protocol rather than a canonical raw-sequence leaderboard, the model-selection procedure is stated explicitly. Prior benchmark studies have shown that both baseline quality and model ranking can change materially once methods are re-evaluated under a common framework and retuned under a shared protocol [21, 22]. Accordingly, the comparison does not rely on paper defaults alone.

Our reporting protocol follows three simple rules. First, every model is tuned within a predefined bounded search space that is fixed before the final comparison: learning rate is sampled log-uniformly within a declared dataset-specific interval, while the remaining hyperparameters are drawn from small predeclared candidate sets. Second, these bounded spaces cover a shared core of optimization hyperparameters together with a small number of model-specific controls, so that the search is broad enough to avoid trivial under-tuning but still budgeted tightly enough to remain comparable across methods. Third, the final reported result for every model is the test performance of the configuration that maximizes the mean validation score over the nine reported metrics. When adaptive search is used for computational efficiency, it is restricted to the same predeclared bounds and candidate sets and should therefore be read as an efficient implementation of a controlled bounded search rather than as open-ended retuning [9].

The final paper track therefore uses bounded search spaces rather than unconstrained retuning. In the paper-facing comparison, baselines and RouteRec are aligned to the same shared optimization pool, and model families differ only through a small number of architecture-specific additions. For RouteRec, the selected three-stage routing layout remains fixed, but depth, hidden dropout, and routing-side controls are still tuned within bounded candidate sets. To keep the appendix concise but still reproducible, Table 6 separates the shared optimization pool from the baseline-family and RouteRec-specific additions, and lists the dataset-specific learning-rate intervals explicitly.

B.3 Full Ranking Results

Table 7 restores the full cutoff grid summarized more compactly in Table 1. This presentation preserves the same selection rule as the main text while exposing the complete $k \in \{5, 10, 20\}$ breakdown.

The appendix table confirms that the same qualitative ordering persists across cutoffs: RouteRec is strongest overall, SASRec and TiSASRec remain robust reference points, and the relatively weaker BSARec and DIF-SR results are consistent with the sessionized short-prefix regime discussed in Section 5.2 rather than with a change in selection or evaluation policy.

This table makes it possible to verify whether the observed trends persist beyond the compact main-text summary under the same selection rule.

Table 6: Bounded candidate grids used in the final paper track.

(a) Shared Optimization Pool

Parameter	Candidate sets
Learning rate	Dataset-specific log-uniform interval; see panel (d)
Weight decay	$\{5 \times 10^{-7}, 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 10^{-4}, 1.5 \times 10^{-4}\}$
Max history length	$\{10, 20, 30\}$ or $\{10, 20, 30, 50\}$
Width / embedding	$\{64, 96, 112, 128, 160\}$ or $\{96, 112, 128, 160, 192\}$
Inner width	$\{128, 192, 224, 256, 320\}$ or $\{192, 224, 256, 320, 384\}$
Layers / heads	Layers $\{1, 2, 3, 4\}$; heads $\{1, 2, 4, 8\}$ if applicable
Dropout	Hidden / ratio: $\{0.10, 0.12, 0.13, 0.15, 0.18\}$ Recurrent: $\{0.10, 0.15, 0.20, 0.25, 0.30\}$ Attention: $\{0.06, 0.08, 0.10, 0.12, 0.15\}$ or $\{0.08, 0.10, 0.12, 0.15, 0.20\}$

(b) Baseline-Family Additions

Model(s)	Additional candidate sets
TiSASRec	Time span $\{64, 128, 256, 384, 512\}$
GRU4Rec	Recurrent dropout $\{0.10, 0.15, 0.20, 0.25, 0.30\}$
DuoRec / FEARec	$\tau\{0.16, 0.18, 0.20, 0.22, 0.24\}$ Contrastive weight $\{0.02, 0.03, 0.04, 0.05, 0.06\}$ Semantic weight $\{0, 0.04, 0.05, 0.08, 0.10\}$ or $\{0.04, 0.05, 0.08, 0.10, 0.12\}$
BSARec	$\alpha\{0.35, 0.50, 0.55, 0.70\}$ $c\{2, 3, 5, 7\}$
DIF-SR / FDSA	Attribute width $\{96, 128, 160, 192\}$ λ_{attr} $\{0.08, 0.09, 0.10, 0.12, 0.14\}$ or $\{0.09, 0.10, 0.12, 0.14, 0.15\}$
DIF-SR	Fusion $\{\text{gate}, \text{sum}, \text{concat}\}$
FAME	Expert count $\{2, 3, 4, 5, 6\}$

(c) RouteRec-Specific Additions

Parameter	Candidate sets
Backbone depth / dropout	Depth: $\{1, 2, 3\}$ Dropout: $\{0.10, 0.12, 0.14, 0.15, 0.16, 0.17\}$ $\{0.18, 0.19, 0.20, 0.22, 0.24\}$
Weight decay	$\{3.125, 5, 6, 10, 15, 20, 50\} \times 10^{-7}$
Routing capacity	Expert scale $\{2, 3, 4\}$ or $\{2, 3, 4, 5\}$ for KuaiRec router width $\{32, 64, 96, 128\}$
Routing dropout	Feature $\{0, 0.03, 0.05, 0.10\}$ attention $\{0.05, 0.06, 0.08, 0.10, 0.12\}$
Regularization	$\lambda_{\text{cons}}\{0, 2.5, 5, 8, 12\} \times 10^{-4}$ $\lambda_z\{0, 5, 10, 20\} \times 10^{-5}$
Cue bank	Cue bank size $\{10, 12, 16, 24\}$ router inner width per cue family (small preset)

(d) Dataset-Specific Learning-Rate Intervals

Dataset	Unified log-uniform LR interval
Beauty	1.5e-4 ∼ 2e-3
Foursquare	1.5e-4 ∼ 2.2e-3
KuaiRec	3e-4 ∼ 5e-3
LastFM	8e-5 ∼ 1.2e-3
ML-1M	1.5e-4 ∼ 2.2e-3
Retail Rocket	1.5e-4 ∼ 2.2e-3

Table 7: Full seen-target experimental results across all cutoffs. Datasets marked with † are sampled processed subsets. For each method, we report the seen-target test metrics of the run achieving the highest mean seen-target test score over the nine reported metrics. Avg. Rank is computed over all rows in this table, where lower is better. The RouteRec column is shaded for readability.

Dataset	Metric	SASRec	GRU4Rec	TiSASRec	FEARec	DuoRec	BSARec	FAME	DIF-SR	FDSA	RouteRec
Beauty	HR@5	0.1038	0.0573	0.0918	0.1003	0.0993	0.0630	0.0372	0.0993	0.1041	**0.1087**
	HR@10	0.1324	0.0840	0.1235	0.1404	0.1347	0.0726	0.0497	0.1165	0.1318	**0.1561**
	HR@20	0.1637	0.1146	0.1630	0.1757	0.1958	0.0955	0.0592	0.1385	0.1681	**0.2009**
	NDCG@5	0.0707	0.0372	0.0729	0.0669	0.0678	0.0474	0.0263	0.0756	**0.0830**	0.0798
	NDCG@10	0.0800	0.0457	0.0831	0.0797	0.0793	0.0504	0.0303	0.0811	0.0918	**0.0948**
	NDCG@20	0.0879	0.0533	0.0931	0.0889	0.0948	0.0563	0.0327	0.0867	0.1011	**0.1062**
	MRR@5	0.0598	0.0307	0.0666	0.0556	0.0571	0.0422	0.0228	0.0677	**0.0761**	0.0701
	MRR@10	0.0637	0.0341	0.0707	0.0608	0.0619	0.0434	0.0244	0.0700	**0.0796**	0.0762
	MRR@20	0.0659	0.0361	0.0735	0.0634	0.0662	0.0450	0.0250	0.0716	**0.0822**	0.0794
Retail Rocket	HR@5	0.4610	0.4312	0.4433	0.4730	0.4711	0.4411	0.4398	0.4262	0.4579	**0.4862**
	HR@10	0.5321	0.5048	0.5131	0.5487	0.5478	0.4655	0.4612	0.4322	0.5343	**0.5608**
	HR@20	0.5988	0.5719	0.5810	0.6234	0.6194	0.4887	0.4812	0.4365	0.6053	**0.6315**
	NDCG@5	0.3676	0.3436	0.3528	0.3752	0.3745	0.3877	0.3874	0.3844	0.3600	**0.3907**
	NDCG@10	0.3907	0.3674	0.3755	0.3998	0.3994	0.3957	0.3943	0.3863	0.3848	**0.4148**
	NDCG@20	0.4076	0.3844	0.3926	0.4187	0.4175	0.4015	0.3994	0.3874	0.4028	**0.4327**
	MRR@5	0.3365	0.3145	0.3227	0.3427	0.3423	0.3696	0.3695	**0.3698**	0.3274	0.3588
	MRR@10	0.3461	0.3243	0.3321	0.3529	0.3526	**0.3729**	0.3724	0.3706	0.3377	0.3688
	MRR@20	0.3507	0.3290	0.3368	0.3581	0.3576	**0.3745**	0.3738	0.3709	0.3427	0.3737
Foursquare	HR@5	0.2350	0.1741	0.2388	0.2403	0.2438	0.1995	0.1954	0.2458	0.2514	**0.2559**
	HR@10	0.3049	0.2180	0.2976	0.3054	0.3195	0.2383	0.2417	0.3032	0.3042	**0.3215**
	HR@20	0.3635	0.2593	0.3501	0.3620	0.3760	0.2773	0.2800	0.3536	0.3524	**0.3776**
	NDCG@5	0.1703	0.1334	0.1735	0.1714	0.1726	0.1483	0.1455	0.1821	**0.1843**	0.1829
	NDCG@10	0.1929	0.1476	0.1926	0.1925	0.1970	0.1609	0.1604	0.2008	0.2014	**0.2042**
	NDCG@20	0.2078	0.1580	0.2059	0.2068	0.2114	0.1708	0.1701	0.2135	0.2136	**0.2184**
	MRR@5	0.1489	0.1199	0.1519	0.1486	0.1491	0.1313	0.1289	0.1611	**0.1621**	0.1588
	MRR@10	0.1583	0.1258	0.1599	0.1574	0.1591	0.1366	0.1351	0.1688	**0.1692**	0.1676
	MRR@20	0.1624	0.1286	0.1635	0.1613	0.1631	0.1393	0.1378	0.1723	**0.1726**	0.1715
ML-1M	HR@5	0.0858	0.0984	0.0906	0.0906	0.0977	0.0892	0.0968	0.0384	**0.1022**	0.1005
	HR@10	0.1423	0.1456	0.1448	0.1490	0.1555	0.1374	0.1533	0.0669	**0.1606**	0.1568
	HR@20	0.2157	0.2181	0.2220	0.2234	0.2402	0.2018	0.2243	0.1052	**0.2426**	0.2353
	NDCG@5	0.0536	0.0626	0.0582	0.0549	0.0624	0.0568	0.0637	0.0240	**0.0645**	0.0627
	NDCG@10	0.0717	0.0778	0.0756	0.0736	0.0810	0.0723	0.0818	0.0331	**0.0833**	0.0807
	NDCG@20	0.0902	0.0960	0.0950	0.0923	0.1022	0.0884	0.0996	0.0427	**0.1038**	0.1004
	MRR@5	0.0431	0.0510	0.0476	0.0432	0.0508	0.0463	**0.0529**	0.0192	0.0522	0.0504
	MRR@10	0.0505	0.0572	0.0548	0.0508	0.0584	0.0527	**0.0603**	0.0230	0.0600	0.0577
	MRR@20	0.0555	0.0621	0.0600	0.0559	0.0641	0.0570	0.0651	0.0256	**0.0655**	0.0631
LastFM†	HR@5	0.3471	0.2836	0.3405	0.3630	0.3521	0.3312	0.3173	0.3389	0.3494	**0.3693**
	HR@10	0.3775	0.3109	0.3715	0.3876	0.3850	0.3475	0.3359	0.3593	0.3762	**0.3965**
	HR@20	0.3999	0.3338	0.3997	0.4103	0.4101	0.3615	0.3554	0.3825	0.4020	**0.4194**
	NDCG@5	0.2999	0.2588	0.2929	0.3108	0.2959	0.2992	0.2911	0.3103	0.3161	**0.3218**
	NDCG@10	0.3097	0.2676	0.3029	0.3187	0.3066	0.3044	0.2972	0.3169	0.3247	**0.3307**
	NDCG@20	0.3154	0.2735	0.3101	0.3245	0.3129	0.3080	0.3020	0.3228	0.3313	**0.3365**
	MRR@5	0.2841	0.2506	0.2770	0.2933	0.2771	0.2884	0.2822	0.3008	0.3050	**0.3059**
	MRR@10	0.2881	0.2542	0.2812	0.2965	0.2815	0.2906	0.2847	0.3035	0.3085	**0.3096**
	MRR@20	0.2897	0.2558	0.2831	0.2981	0.2833	0.2915	0.2861	0.3051	0.3103	**0.3113**
KuaiRec†	HR@5	0.3195	0.2770	0.3004	0.3397	0.3270	0.3281	0.3255	0.2275	0.3270	**0.3494**
	HR@10	0.3284	0.3061	0.3216	0.3513	0.3468	0.3359	0.3307	0.2333	0.3453	**0.3669**
	HR@20	0.3379	0.3367	0.3429	0.3688	0.3677	0.3464	0.3430	0.2411	0.3654	**0.3949**
	NDCG@5	0.3148	0.2493	0.2820	0.3249	0.3152	0.3184	0.3172	0.2210	0.3140	**0.3410**
	NDCG@10	0.3177	0.2589	0.2889	0.3286	0.3215	0.3209	0.3188	0.2229	0.3197	**0.3467**
	NDCG@20	0.3200	0.2666	0.2942	0.3330	0.3269	0.3236	0.3220	0.2248	0.3248	**0.3538**
	MRR@5	0.3133	0.2400	0.2759	0.3200	0.3112	0.3152	0.3144	0.2188	0.3097	**0.3382**
	MRR@10	0.3144	0.2441	0.2788	0.3215	0.3138	0.3162	0.3151	0.2196	0.3120	**0.3406**
	MRR@20	0.3150	0.2462	0.2802	0.3227	0.3153	0.3170	0.3160	0.2201	0.3133	**0.3425**
Avg. Rank↓		**6.07**	**8.63**	**6.56**	**4.41**	**4.41**	**6.52**	**6.72**	**6.55**	**3.40**	**1.74**

C Where RouteRec Helps Most: Special Bins

This section examines coarse difficulty-oriented bins such as session length, target frequency, and optional user history, with the goal of showing where the average RouteRec gain is concentrated. The reported analyses distribute gains across session-length, item-frequency, and optional user-history regimes rather than treating the average result as a black box, making the Q1 improvement pattern more interpretable before the paper turns to architectural justification and sparse-routing design.

Table 8: Special-bin definitions and sample counts used in the appendix gain analysis.

Bin family	Definition rule	Visualization
Session length	Short / medium / long session buckets from the observed prefix length	Figure 6(a)
Target frequency	Rare / mid / frequent target buckets under the train split	Figure 6(b)
User history	Optional low-history / medium-history / heavy-history grouping	Supporting figure

These figures are positioned before the structural ablations to show that Q1 is not merely an average-score claim: RouteRec produces its clearest margin in behaviorally heterogeneous or otherwise difficult regimes, which motivates the later appendix sections on design and routing semantics.

D Extended Framework and Structural Ablations

This section extends the main-text design-justification study by examining broader temporal-layout and cue-organization variants within the RouteRec framework. Figure 7 collects the family-grouping controls and temporal-layout variants that are too detailed for the main text, while preserving the same Q3 logic: the selected paper architecture should look like a stable design choice rather than an arbitrary expertized backbone.

Panel (a) tests whether semantic grouping matters beyond a flat scalar bag. Panel (b) checks whether the macro, mid, and micro roles correspond to meaningfully distinct temporal views rather than just repeated copies of the same router, and whether the selected stage ordering remains the most stable high-performing layout among the temporal controls.

E Sparse Routing Design Variants

Table 9 tests whether the selected sparse hierarchy is an arbitrary engineering choice or a meaningful part of the RouteRec design. This section directly supports Q3 and Q4 by contrasting dense routing, flat sparse controls, and hierarchical top-k variants under comparable activation budgets.

Table 9: Sparse routing design variants, ranging from a dense reference to progressively sparser hierarchical configurations under comparable total expert pools.

Variant	MRR@20	Active experts	Observation
Dense full mixture	0.336	12	Reference control
Flat sparse top-6	0.335	6	Same active count without hierarchy
Top-2gr, top-1ex	0.332	2	Extremely sparse reference
Top-2gr, top-2ex	0.335	4	Strong group commitment, tight capacity
Top-3gr, top-2ex	**0.339**	6	**Selected main setting**
Top-4gr, top-2ex	0.337	8	Weaker group commitment
Top-3gr, top-3ex	0.338	9	Larger active-compute budget reference

The table separates three related questions: whether sparsity itself helps, whether hierarchy matters beyond a flat top-k mask, and whether the selected main setting provides a reasonable balance between route commitment and active capacity.

F Objective and Regularization Variants

This table isolates whether the observed routing semantics depend primarily on the selected auxiliary objective or remain visible across objective variants. Table 10 reports ranking quality together with routing-stability indicators for each objective configuration, allowing the diagnostics in Appendix H to be read with the objective contribution accounted for.

Table 10: Objective and regularization variants.

Variant	MRR@20	Neighbor JSD ↓	Route-entropy var. ↓
No auxiliary loss	0.333	0.148	0.094
KNN consistency only	0.336	0.127	0.091
z-loss only	0.337	0.136	0.083
Balance loss only	0.334	0.141	0.090
Consistency + z-loss	0.338	0.124	0.080
Full objective	**0.339**	**0.121**	**0.079**

The two central comparisons are whether the selected objective improves ranking under sparse routing, and whether removing specific auxiliary terms materially changes the sharpness and consistency patterns reported in Appendix H.

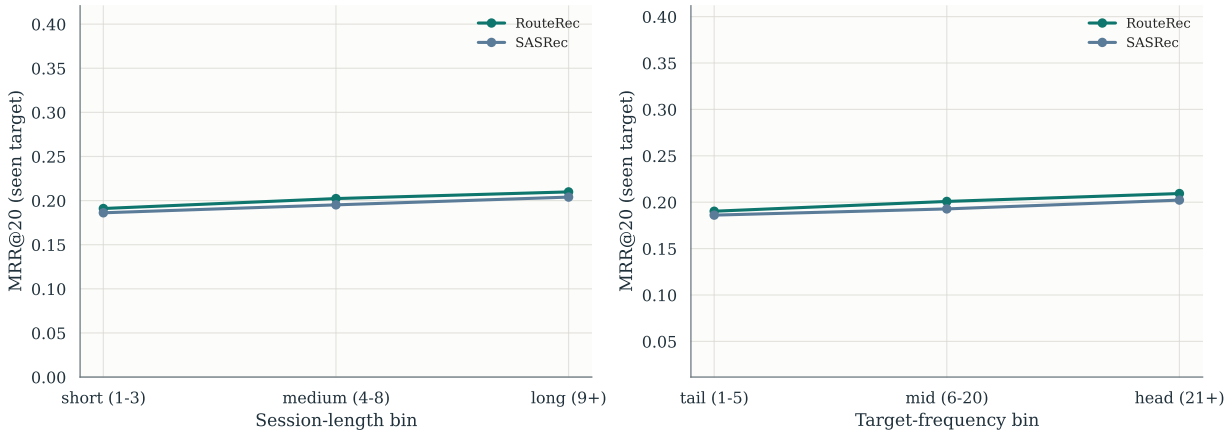
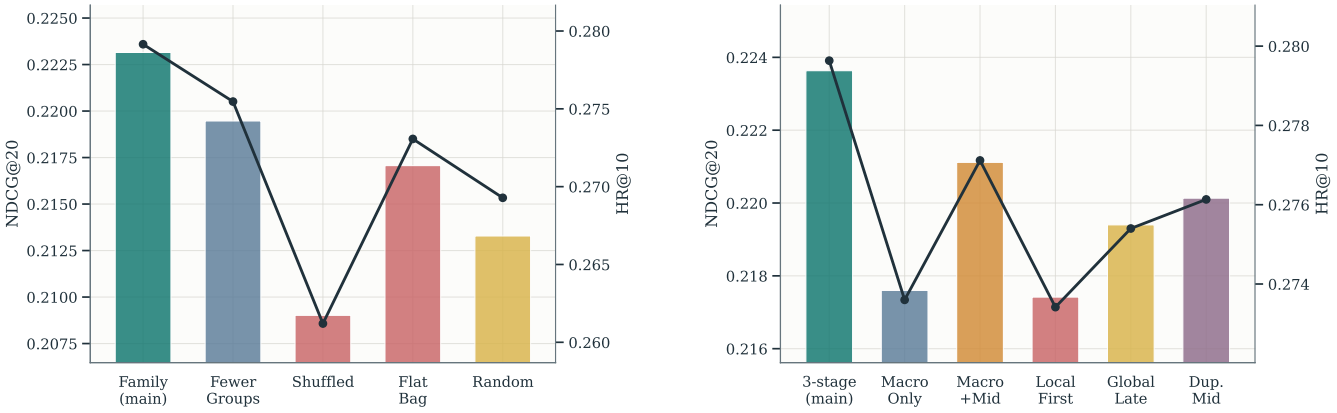


Figure 6: Special-bin analysis reporting where the average RouteRec gain is concentrated across session-length and target-frequency regimes. These bins are coarse difficulty- and exposure-oriented partitions rather than semantically defined behavioral regimes; those behavior-centric slices are examined separately in Appendix I.



(a) Cue and family-group organization variants

(b) Stage-semantics and temporal-layout variants

Figure 7: Extended structural ablations. (a) Semantic cue and family-group organization variants. (b) Stage-semantics, temporal-role controls, and layout variants.

G Full Cost and Efficiency Summary

Table 11 provides a compact cost summary for the claim that the routing branch is lightweight. Rather than turning the appendix into a systems study, this section reports the rough parameter and runtime overhead for representative baselines together with RouteRec's active-parameter view under sparse routing.

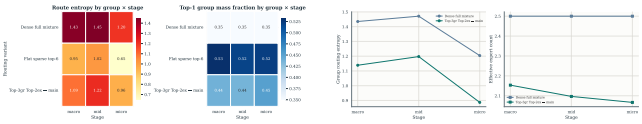
Table 11: Model cost and efficiency summary for all compared methods.

Model	Params	Active params	Train ×	Infer ×	Note
GRU4Rec	1.4	1.4	0.83×	0.88×	Dense recurrent reference
SASRec	1.8	1.8	1.00×	1.00×	Normalization anchor
TiSASRec	2.1	2.1	1.31×	1.22×	Time-aware Transformer
DuoRec	2.0	2.0	1.28×	1.15×	Contrastive sequential
FEARec	2.0	2.0	1.35×	1.24×	Frequency-aware
BSARec	1.9	1.9	1.18×	1.10×	Structural sequential
FAME	2.8	2.8	1.62×	1.48×	Expert-based baseline
DIF-SR	2.4	2.4	1.42×	1.32×	Feature-aware sequential
FDSA	2.3	2.3	1.38×	1.28×	Feature-side attention
RouteRec	3.2	1.6	1.42×	1.18×	Active params: routed subset only

A compact cost comparison is sufficient here to anchor the selective-activation claim and to show that the observed gains are not being attributed solely to an unchecked increase in compute. The timing numbers in this table are measured on a single NVIDIA RTX 3090 24 GB GPU under the same batch-size and evaluation conditions; the ratios serve as a practical reference point rather than as a deployment-grade systems benchmark.

H Extended Routing Diagnostics

This section summarizes the final model's routing structure through aggregate diagnostics rather than controlled perturbations; targeted interventions are deferred to Appendix J. Figure 8 groups the main diagnostics so that group usage, route sharpness, and stage-wise consistency can be read together. Together with the main-text cases, these plots provide the aggregate support for Q5.



(a) Stage-wise family and expert usage heatmaps

(b) Route entropy and effective-expert-count summary

Figure 8: Routing diagnostics. (a) Group/expert usage heatmaps by stage. (b) Route entropy and effective expert count across variants. Targeted intervention results are reported separately in Appendix J.

These diagnostics are organized from coarse structure to semantic interpretation. Panel (a) shows how family and expert usage is distributed by stage, while Panel (b) summarizes route sharpness and effective expert usage across the main sparse-routing variants.

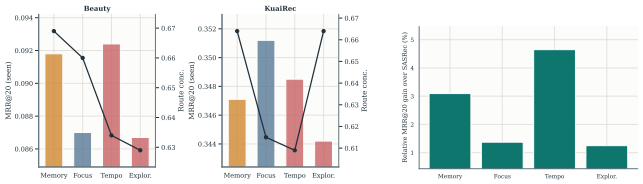
I Extended Behavioral Slice Results

Unlike Appendix C, which analyzes coarse difficulty-oriented bins, this section examines semantically defined behavioral regimes such as repeat-heavy, fast exploratory, narrow-focus, and preference-stable slices. It first defines the diagnostic slices explicitly, then extends the main-text regime analysis with per-dataset breakdowns and more detailed routing summaries.

Table 12: Diagnostic slice definitions used in the behavior-regime analysis.

Slice	Definition rule
Repeat-heavy	Prefix repeat ratio above a fixed percentile threshold
Fast-tempo	Mean inter-event gap below a fixed percentile threshold
Narrow-focus	Category or item-transition entropy below a fixed percentile threshold
Exploration-heavy	Novel-item or low-repeat ratio above a fixed percentile threshold

These slices are used only for post hoc diagnosis. They are computed from simple observable statistics of the interaction history, not from optimized supervision targets, and not from labels learned jointly with the model.



(a) Slice-wise MRR@20 across behavioral regimes

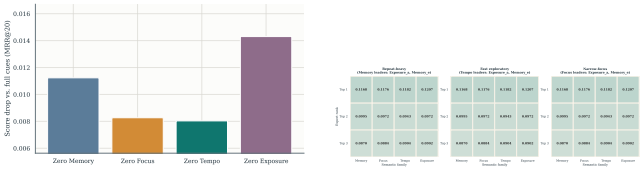
(b) Relative RouteRec gain by slice

Figure 9: Extended behavior-regime analysis. (a) Slice-wise ranking quality across behavioral regimes. (b) Relative RouteRec gain over SASRec by behavioral slice.

Together, the two panels expose dataset-level heterogeneity that would be too detailed for the main paper while making the slice construction explicit enough to avoid circular interpretation.

J Targeted Interventions and Qualitative Cases

This section complements the main-text case visualizations with controlled cue-family perturbations and directly interpretable qualitative examples. The main text presents three representative cases; the appendix adds a preference-stable contrastive case and the matched intervention evidence. Together, the intervention panels and the case table make the Q5 semantic claim readable at both the distribution level and the level of individual sessions.



(a) Score drop by cue-family perturbation

(b) Stage-wise routing profiles by behavioral case

Figure 10: Targeted intervention analysis. (a) Score drop under cue-family perturbation for each behavioral regime. (b) Stage-wise routing profiles for the representative behavioral cases.

Table 13: Representative qualitative routing cases. Selected groups and experts are post hoc observations from the final model's routing decisions on each case type.

Case	Prefix summary	Selected groups	Prediction outcome	Interpretation
Repeat-heavy	High repeat ratio; low inter-event variance; stable category distribution	Memory (macro, micro); Tempo (macro)	Correct revisit prediction; MRR≈0.41	Macro and micro routes reinforce revisit behavior
Fast exploratory	Short inter-event gaps; high switch rate; low category entropy	Tempo (micro); Exposure (macro, mid)	Reasonable exploration target; MRR≈0.35	Micro routing sharpens under short-horizon volatility
Narrow-focus	High dominant-group mass; low entropy; few distinct categories	Focus (mid, micro); Memory (macro)	Category-consistent recommendation; MRR≈0.39	Mid-stage routing concentrates on coherent intent
Preference-stable	Long history; low recency drift; uniform pace	All groups near uniform; low route entropy	Moderate improvement over SASRec	Gains narrow when routing demand is weak

The intervention panels explain which semantic families the router appears to rely on most, while Table 13 complements that distribution-level view with directly interpretable cases.

K Optional Transfer Results

This section reports optional transfer-style results examining whether RouteRec's portable routing interface can be reused across domains with different behavioral and temporal characteristics. The source-target matrix and low-resource curves provide the pairwise and data-efficiency views omitted from the main text.

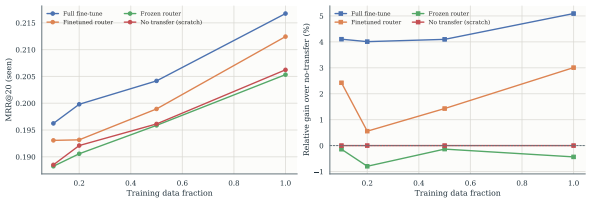


Figure 11: Transfer analysis. Low-resource MRR@20 curves across data fractions comparing full fine-tune, finetuned router, anchor transfer, frozen router, and no-transfer baselines.

Figure 11 shows how the routing interface transfers across data-efficiency regimes. Full fine-tuning and finetuned-router transfer consistently outperform the no-transfer baseline, while the frozen-router setting remains a lightweight alternative at higher data fractions.